A Mathematical Primer for Wald and Ornstein–Uhlenbeck Race Models

August 27, 2026

Abstract

This note gives a mathematical description of one-boundary Wald and Ornstein–Uhlenbeck (OU) accumulators, their racing constructions (including the Race of OUs, or ROU, and the pooled/coactivation ROUp model), bounded two-boundary OU diffusion (BOU), the pulse-drift extension, and a finite time transformation of the Wald process (RDMSWTN_TT). It also collects identities for non-decision time, uniform start-point variability, Gaussian drift variability, Gaussian copulas for correlated finishing times, and Erlang guess and kill clocks, including a hybrid clock that mixes shape 1 and shape 2 timing. Special attention is given to defective negative-drift Wald processes, the distinction between full-Gaussian and positive-truncated drift laws (and how they contrast with rectified models), copula rank coupling of finite and defective finishing times, and the boundary between closed-form expressions and one-dimensional quadrature. Ballistic race models are not covered here; they are derived in a companion note, since they are not first-passage laws and share none of the machinery below.

Contents

1 Theory Map and Reading Guide

The model combines a few standard mathematical objects. The terminology follows the supplied chapters from Smith and Ratcliff's *Diffusion Process Models of Decision Making*: the sequential-sampling overview (Chapter 3, especially Sections 3.6–3.8) and the treatment of time-varying diffusion models for detection and simple RT (Chapter 8, especially Sections 8.2–8.5). The corresponding chapter records are linked here: sequential-sampling chapter and time-varying-diffusion chapter. The supplied PDF extracts in `Math/` are the working copies used for this note.

- Brownian motion and stopping times: the general language is Brownian motion and hitting time. In the sequential-sampling terminology, the Wiener process is a continuous-time, continuous-state diffusion whose infinitesimal drift and diffusion rates are the counterparts of the mean and variance of random-walk increments (Smith & Ratcliff, Chapter 3, Sections 3.6–3.8). The first-passage time to an absorbing criterion is a stopping time, which is why Laplace-transform and optional-stopping arguments are natural. The Wald process below is the one-boundary Wiener special case used throughout that chapter.

- The Wald or inverse-Gaussian law: see the inverse Gaussian distribution. In first-passage parameterization, the density below is the hitting-time density of a drifted Brownian motion crossing a fixed boundary.

- The Ornstein–Uhlenbeck process: the OU diffusion is a Brownian motion with linear mean reversion (leak), the state-dependent drift example highlighted in Chapter 3. Its first-passage law is defined by the killed transition density and boundary flux; the constant-drift Wald process is recovered when the leak is zero. Chapter 8 uses single-boundary OUs driven by filtered sustained and transient channels. In that terminology, ROU is the race between those independent OU detectors, whereas ROUp is the pooled (coactivation) alternative in Smith (1995), with one accumulator receiving the sum of the channel drives. The BOU model generalizes this to a two-boundary OU diffusion process.

- Time-varying evidence: Chapter 8 emphasizes that stimulus encoding generally produces smooth drift functions $\mu(t)$ and, potentially, time-varying diffusion rates rather than a step change. Its sustained channel rises toward an asymptote and its transient channel rises to a peak before decaying. Our pulse-drift extension is a deliberately simpler parametric forcing; the ROU and ROUp labels identify the corresponding race and pooled architectures, not a claim that every implementation reproduces Smith's filter cascade.

- The reflection-principle idea: the two terms in the Wald CDF come from the reflection principle. The second term is often called the image term.

- Gaussian integration identities: univariate normal functions are summarized in the NIST normal distribution reference. Bivariate-normal probabilities are summarized by MathWorld's bivariate normal page and the broader multivariate normal distribution theory.

- Gaussian copula and rank dependence: Sklar's theorem decomposes joint hitting-time distributions into marginal CDFs and a copula function; see copula theory and Sklar's theorem. The Gaussian (probit) copula maps marginal uniform ranks to standard normal variates via Φ^{-1}. For defective marginal distributions ($F(\infty) < 1$), copula ranks above the marginal upper bound represent the point mass at infinity, coupling intrinsic omissions as well as finite finishing times.

- Erlang timing: Erlang clocks are integer-shape gamma clocks; see MathWorld's Erlang distribution page. Shape 1 is exponential timing. Shape 2 adds one polynomial factor to the exponential survival. The hybrid shape 3 process used below is not an ordinary Erlang-3 clock; it is a two-component mixture of shape 1 and shape 2 clocks.

The most important conceptual split in this note is between *retaining negative drifts* and *truncating the drift distribution at zero*. Retaining negative drifts gives a defective first-passage distribution. Truncating at zero gives a proper positive-drift distribution, but it changes the drift law itself.

For the historical model anchors, Smith's one-choice visual model is Smith (1995), *Psychophysically Principled Models of Visual Simple Reaction Time*; the one-choice diffusion model and its zero-or-positive drift comparison are Ratcliff and Van Dongen (2011). These references are used below for model correspondence, not as substitutes for the derivations given here.

2 Time Domains and Basic Wald Law

Let T denote raw response time and let

$$x = T - t_0$$

be evidence-accumulation time. The evidence process starts at t_0, so evidence-response density is zero for $x \leq 0$. Timing clocks such as guess and kill processes start at raw trial time zero unless stated otherwise. This separation follows the sequential-sampling convention: the first-passage calculation is carried out in decision time, and non-decision time is added afterward.

We use d for the remaining distance to a one-sided criterion. To translate to the notation used in the canonical sequential-sampling treatments, read d as the criterion distance (often a), μ or m as the mean drift rate, and σ^2 as the diffusion rate; s_v is the across-trial standard deviation of the drift rate (often written η). A random start point is equivalent to a random distance, $D = a - Z_0$ (up to the choice of origin), so our range A is the start-point range expressed in distance units.

For a fixed distance $d > 0$, drift μ, and diffusion coefficient $\sigma > 0$, define

$$X_t = \mu t + \sigma W_t, \qquad \tau_d = \inf\{t > 0 : X_t = d\}.$$

This is the one-boundary Wiener/Brownian diffusion: its infinitesimal drift is μ and its diffusion rate is σ^2. The first-passage density is

$$f_W(x; d, \mu, \sigma) = \frac{d}{\sqrt{2\pi\sigma^2 x^3}} \exp\left[-\frac{(d-\mu x)^2}{2\sigma^2 x}\right], \qquad x > 0. \tag{1}$$

Its sub-CDF is

$$F_W(x; d, \mu, \sigma) = \Phi\left(\frac{\mu x - d}{\sigma\sqrt{x}}\right) + \exp\left(\frac{2\mu d}{\sigma^2}\right) \Phi\left(-\frac{d + \mu x}{\sigma\sqrt{x}}\right). \tag{2}$$

For $\mu < 0$ this is defective:

$$F_W(\infty; d, \mu, \sigma) = \exp\left(\frac{2\mu d}{\sigma^2}\right) < 1. \tag{3}$$

For $\mu \geq 0$, $F_W(\infty; d, \mu, \sigma) = 1$. Thus negative drift is not impossible; it simply induces a positive non-hitting probability.

Uniform start-point variability is represented as a random distance

$$D \sim U[b - A, b], \qquad 0 \leq A < b. \tag{4}$$

The start-point averaged CDF is

$$F_A(x) = \frac{1}{A} \int_{b-A}^{b} F_W(x; d, \mu, \sigma) \, \mathrm{d}d, \tag{5}$$

with the obvious point-mass limit when $A = 0$.

3 Gaussian Drift Variability

Let the across-trial drift be

$$V \sim \mathcal{N}(m, s_v^2). \tag{6}$$

We use "between-trial" and "across-trial" interchangeably for this trialwise draw; both distinguish it from the within-trial Brownian noise. Two drift laws are useful and should not be conflated.

The reason drift variability can sometimes be integrated analytically is that, conditional on $V = v$, the accumulator value at a fixed time is Gaussian with mean vx and variance $\sigma^2 x$. Mixing v over a Gaussian distribution preserves Gaussianity at fixed time. The complication is that a first-passage CDF is not merely the fixed-time crossing probability; it also contains the image term from the reflection principle.

3.1 Full-Gaussian Drift Law

Under the full-Gaussian law, negative drifts are retained. They contribute finite hit density through the defective Wald law and non-hit mass through $1 - F_W(\infty; d, V, \sigma)$. Note that the derivation for between-trial variability without truncation is mathematically the same (though using different symbology) as the derivation given by Smith (1995) for the coactivation model, but here we extend it to consider start point variability ($A > 0$, derived in a subsequent section) and we derive the exact CDFs as well.

More explicitly, the shared step is the Gaussian mixing of a one-boundary first-passage law over a trialwise drift: conditional on $V = v$, use the Wald density or CDF, then take expectation under a Gaussian drift distribution. In Smith's notation this is the same operation as mixing over the across-trial drift parameter (with the criterion, mean drift, diffusion rate, and drift SD corresponding here to d, m, σ^2, and s_v). The change of symbols and our explicit evidence-time variable x do not change that identity. The extensions in this note are (i) averaging over a uniform random distance $D \sim U[b - A, b]$, which is the start-point variability written in distance coordinates, (ii) retaining the image term when deriving the CDF rather than only the density, and (iii) recording the resulting finite hit mass and atom at infinity. Thus the no-start-variability density in (7) is the shared Smith-style calculation, while the CDFs and the $A > 0$ results below are the present extension.

For fixed d and $s_v > 0$, integrating the Wald density over the full normal drift distribution gives

$$f_{\mathrm{FG}}(x; d) = \frac{d}{\sqrt{2\pi x^3(\sigma^2 + s_v^2 x)}} \exp\left[-\frac{(d - mx)^2}{2x(\sigma^2 + s_v^2 x)}\right], \qquad x > 0. \tag{7}$$

The corresponding fixed-criterion CDF has a simple univariate-normal form. Define

$$Q(x) = \sigma^2 + s_v^2 x, \qquad h_1(x) = \frac{mx - d}{\sqrt{xQ(x)}}, \tag{8}$$

$$\alpha = \frac{2d}{\sigma^2}, \qquad m_+ = m + \alpha s_v^2, \qquad h_2(x) = \frac{-m_+ x - d}{\sqrt{xQ(x)}}. \tag{9}$$

Then

$$F_{\mathrm{FG}}(x; d) = \Phi(h_1(x)) + \exp\left(\alpha m + \frac{1}{2}\alpha^2 s_v^2\right) \Phi(h_2(x)). \tag{10}$$

For a nondegenerate Gaussian drift law, the negative tail makes this mixture generally defective (the defect may be numerically negligible when m/s_v is large). Its eventual hit probability is

$$H_{\mathrm{FG}}(d) = \Phi\left(\frac{m}{s_v}\right) + \exp\left(\alpha m + \frac{1}{2}\alpha^2 s_v^2\right) \Phi\left(-\frac{m + \alpha s_v^2}{s_v}\right). \tag{11}$$

Equation (10) is the simplification enabled by retaining negative drift and treating non-hits as genuine defective mass rather than conditioning them away.

One way to see (10) is to use two elementary Gaussian identities. If $Z \sim \mathcal{N}(0, 1)$, then

$$E\left[\Phi(a + bZ)\right] = \Phi\left(\frac{a}{\sqrt{1 + b^2}}\right). \tag{12}$$

The first term in (2) follows directly from (12). The image term also has the exponential tilt $\exp(\alpha V)$ with $\alpha = 2d/\sigma^2$. Exponential tilting of a Gaussian shifts its mean:

$$e^{\alpha v}\varphi(v; m, s_v^2) = \exp\left(\alpha m + \frac{1}{2}\alpha^2 s_v^2\right)\varphi(v; m + \alpha s_v^2, s_v^2). \tag{13}$$

Applying (12) again under the tilted Gaussian gives the second term in (10).

3.2 Positive-Truncated Drift Law

Under the positive-truncated law,

$$V_+ \equiv V \mid V > 0, \qquad Z_+ = P(V > 0) = \Phi\left(\frac{m}{s_v}\right). \tag{14}$$

The density of V_+ is therefore

$$g_+(v) = \frac{\varphi(v; m, s_v^2)\,\mathbf{1}\{v > 0\}}{Z_+}. \tag{15}$$

The location parameter m remains the location of the original Gaussian; it is not replaced by the conditional mean $E[V \mid V > 0]$. Positive truncation is a distribution-side operation, not a reparameterization of m.

This positive-drift-truncated RDMSWTN is different from the zero-or-positive comparison fitted by Ratcliff and Van Dongen (2011). Their rectified version starts with $V \sim \mathcal{N}(m, s_v^2)$ and clips each draw,

$$V_{\text{clip}} = \max(V, 0), \qquad P(V_{\text{clip}} = 0) = \Phi\left(-\frac{m}{s_v}\right). \tag{16}$$

Consequently, their law retains the original (unrenormalized) positive Gaussian density and adds a point mass at zero; its mean is $E[\max(V, 0)]$, not the conditional mean under $V > 0$. Our law discards the nonpositive draws, rescales the positive density by $1/Z_+$, and has no atom at zero. The distinction is behavioral as well as notational: the clipped model preserves a zero-drift subpopulation that can produce very long responses, whereas every draw in the present model has strictly positive drift and hence a proper eventual first-passage distribution. A zero-drift Brownian process hits a positive boundary almost surely on an infinite horizon (although its mean hitting time is infinite), so the point mass in the rectified law is not itself an infinite-time omission; under a finite observation or exhaustion window it appears as a long-tail or unresolved component.

For completeness, the original Ratcliff–Van Dongen one-choice fit also considered the unmodified Gaussian drift law. Negative draws then generate non-terminating paths, and their RT distributions were evaluated by a finite random-walk simulation. Our full-Gaussian section keeps those negative drifts but treats their non-hitting probability analytically as defective mass; it is not the clipped comparison model.

This drift-law distinction is separate from the finite clock in RDMSWTN_TT. The latter assigns an explicit operational capacity and therefore an explicit exhaustion mass;

a simulation cutoff used to collect RT quantiles is a numerical observation limit, not a zero-drift component of the generative model.

For fixed d the positive-truncated density is

$$f_+(x;d) = \frac{d}{\sqrt{2\pi x^3(\sigma^2 + s_v^2 x)}} \exp\left[-\frac{(d-mx)^2}{2x(\sigma^2 + s_v^2 x)}\right] \frac{\Phi(\mu_*(x)/s_*(x))}{Z_+}, \tag{17}$$

where

$$\mu_*(x) = \frac{ds_v^2 + m\sigma^2}{s_v^2 x + \sigma^2}, \qquad s_*(x) = \sqrt{\frac{\sigma^2 s_v^2}{s_v^2 x + \sigma^2}}. \tag{18}$$

The no-clock positive-truncated CDF is still closed form, but it uses bivariate-normal rectangle probabilities because the integration domain imposes $V > 0$. Let

$$z_0 = -\frac{m}{s_v}, \qquad \rho(x) = \frac{s_v\sqrt{x}}{\sqrt{\sigma^2 + s_v^2 x}}. \tag{19}$$

With h_1, h_2, α, m_+ as above and $z_0^+ = -m_+/s_v$,

$$F_+(x;d) = \frac{1}{Z_+}\left\{\Phi(h_1) - \Phi_2(z_0, h_1; -\rho) + \exp\left(\alpha m + \frac{1}{2}\alpha^2 s_v^2\right)\left[\Phi(h_2) - \Phi_2(z_0^+, h_2; \rho)\right]\right\}. \tag{20}$$

Because all sampled drifts are positive, $F_+(\infty; d) = 1$. This is not the same as normalizing the full-Gaussian defective CDF by $H_{\mathrm{FG}}(d)$; the latter conditions on eventual hitting, whereas positive truncation changes the drift distribution before the Wald process is sampled.

The bivariate-normal terms in (20) arise because the event $V > 0$ and the latent probit event inside the Wald CDF are jointly Gaussian inequalities. The denominator Z_+ only normalizes the drift distribution. It is not an eventual-hit normalizer, and it should not be replaced by $H_{\mathrm{FG}}(d)$.

4 Uniform Start-Point and Drift Variability

When $D \sim U[b-A, b]$ and $V \sim \mathcal{N}(m, s_v^2)$ are independent, the full-Gaussian no-clock density has a simple closed form. Using

$$f_W(x; d, V, \sigma) = \frac{d}{x}\varphi(d; Vx, \sigma^2 x), \tag{21}$$

define the latent Gaussian variable

$$Y_x = Vx + \sigma\sqrt{x}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1). \tag{22}$$

Then

$$Y_x \sim \mathcal{N}(mx, s_D^2), \qquad s_D^2 = s_v^2 x^2 + \sigma^2 x. \tag{23}$$

Averaging over $D \in [b - A, b]$ gives

$$f^{\mathrm{FG}}_{A,s_v}(x) = \frac{1}{Ax} E\left[Y_x \mathbf{1}\{b - A \le Y_x \le b\}\right]. \tag{24}$$

Equivalently, with

$$a_L = \frac{b - A - mx}{s_D}, \qquad a_H = \frac{b - mx}{s_D}, \tag{25}$$

$$f^{\mathrm{FG}}_{A,s_v}(x) = \frac{mx\left[\Phi(a_H) - \Phi(a_L)\right] + s_D\left[\phi(a_L) - \phi(a_H)\right]}{Ax}. \tag{26}$$

This is the start-point averaged version of (7); no bivariate normal is needed because there is no lower truncation on V.

For the positive-truncated drift law, the same density is a first moment over a truncated bivariate-normal rectangle. Let

$$X = \frac{V - m}{s_v}, \qquad Y = \frac{Y_x - mx}{s_D}, \qquad \rho = \frac{s_v\sqrt{x}}{\sqrt{\sigma^2 + s_v^2 x}}, \qquad \gamma = -\frac{m}{s_v}. \tag{27}$$

Then (Y, X) is standard bivariate normal with correlation ρ, and

$$f^{+}_{A,s_v}(x) = \frac{mxP_R + s_D M_R}{AZ_+ x}, \tag{28}$$

where

$$P_R = \left[\Phi(a_H) - \Phi(a_L)\right] - \left[\Phi_2(a_H, \gamma; \rho) - \Phi_2(a_L, \gamma; \rho)\right] \tag{29}$$

and

$$M_R = \left[-\phi(y)\Phi\left(\frac{\rho y - \gamma}{\sqrt{1 - \rho^2}}\right) + \rho\phi(\gamma)\Phi\left(\frac{y - \rho\gamma}{\sqrt{1 - \rho^2}}\right)\right]_{a_L}^{a_H}. \tag{30}$$

For CDFs with both $A > 0$ and $s_v > 0$, one may average the fixed-criterion CDF over $d \in [b - A, b]$. The full-Gaussian CDF uses the univariate expression (10); the positive-truncated CDF uses (20). The image term contains products of drift and distance after start-point averaging, so the resulting integral is not generally reducible to the same elementary Gaussian identities.

This distinction is useful for intuition. The density is local in time and can be written as a fixed-time Gaussian moment over the interval $[b - A, b]$. The CDF is pathwise: it asks whether the process has ever crossed by time x. The pathwise correction is exactly the image term, and that term is what prevents a comparably simple two-variability CDF.

5 Ornstein–Uhlenbeck and Related Models

The Ornstein–Uhlenbeck (OU) diffusion generalizes the standard Brownian accumulation by introducing a linear mean-reversion or leak term. This is the state-dependent-drift example used to situate OU processes in the sequential- sampling taxonomy. For a fixed distance d, starting point $X_0 = 0$, drift profile $\mu(t)$, diffusion coefficient σ, and leak rate λ (or k), the accumulator satisfies

$$\mathrm{d}X_t = [\mu(t) - \lambda X_t]\ \mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{31}$$

The constant-drift Wald process corresponds to the special case where $\lambda = 0$ and $\mu(t) \equiv \mu$. The time-varying chapter emphasizes that stimulus encoding is generally represented by a smooth drift profile (and can also be allowed to change the diffusion rate), rather than by a step change at stimulus onset. The first-passage law for a time-inhomogeneous OU is therefore defined through its killed transition density and boundary flux, and is normally evaluated numerically. The integral-equation strategy used in the time-varying chapter is the relevant general method when no stationary OU first-passage simplification is available.

Three model architectures build on the OU process, matching the sustained– transient construction in Smith (1995) and the sequential-sampling and time-varying chapters:

1. **ROU (Race of OUs)**: Represents a race between independent single-boundary OU accumulators. For sustained and transient channels, write

$$\mathrm{d}X_j(t) = [\mu_j(t) - \lambda_j X_j(t)]\ \mathrm{d}t + \sigma_j\,\mathrm{d}W_j(t), \qquad j \in \{S, T\}, \tag{32}$$

   with independent channel noises W_S and W_T, and let T_j be the first time X_j reaches its criterion. The ROU decision time is $T_{\mathrm{ROU}} = \min(T_S, T_T)$, exactly the parallel sustained–transient race described in Chapter 8.

2. **ROUp (Pooled / Coactivation)**: The Smith (1995) coactivation model uses a single OU accumulator rather than a race. Its channel drives are pooled before accumulation,

$$\mathrm{d}X_p(t) = [\mu_S(t) + \mu_T(t) - \lambda X_p(t)]\ \mathrm{d}t + \sigma\,\mathrm{d}W(t), \tag{33}$$

   so its decision time is one first-passage time. ROUp is therefore not the minimum of two hitting times, and it should not be interpreted as adding the two ROU response densities after the fact.

3. **BOU (Bounded OU)**: Generalizes the OU process to a two-boundary diffusion (e.g., an absorbing upper boundary for detection and a reflecting lower boundary, or two absorbing boundaries for two-choice tasks), ensuring the evidence accumulation remains bounded.

The notation $\mu_S(t)$ and $\mu_T(t)$ is intentional. In the canonical time-varying account, the sustained profile rises toward an asymptote and the transient profile rises to a peak and then decays, after stimulus encoding has been passed through the corresponding low-pass filters. A simple pulse or piecewise profile used in a particular RDMSWTN implementation is a reduced forcing choice; the architectural distinction between ROU and ROUp is the race-versus-pooling distinction above.

6 Time-Transformed Wald (RDMSWTN_TT)

The Time-Transformed Wald (RDMSWTN_TT) model applies a finite linear exhaustion clock to each ordinary accumulator. Let $x = T - t_0$ be the raw decision time. Rather than accumulating in real time x, the process operates in a transformed operational time $q(x)$.

For a finite exhaustion parameter $\tau > 0$, the operational time is defined as:

$$q(x) = x - \frac{x^2}{2\tau}, \qquad 0 < x < \tau. \tag{34}$$

The maximum accumulated operational time is the capacity (or budget) $Q = \tau/2$, which is reached exactly at $x = \tau$. For $x \geq \tau$, the clock is exhausted and operational time plateaus at Q.

This construction is distinct from the time-varying diffusion models in Chapter 8. Those models change the within-trial drift (and, in general, the diffusion rate) as a function of physical time. Here the parent Wiener process remains stationary in operational time and q is an outer deterministic clock: the finite-window law is a pushforward of the ordinary first-passage law. A nonlinear q can of course be rewritten as a time-dependent effective drift and diffusion in raw time, but the RDMSWTN_TT parameterization is chosen to make the finite capacity and its resulting omission mass explicit.

Consequently, the process probability density is zero at and beyond $T \geq t_0 + \tau$. The cumulative distribution function remains fixed at the ordinary CDF evaluated at Q:

$$F_{\mathrm{TT}}(x) = \begin{cases} F_W(q(x)), & 0 < x < \tau \\ F_W(Q), & x \geq \tau \end{cases} \tag{35}$$

The remaining probability mass, $1 - F_W(Q)$, represents genuine omission mass resulting from the exhaustion of the clock before the criterion could be reached. In the limit as $\tau \to \infty$, the operational time reduces to the exact identity clock $q(x) = x$, recovering the ordinary RDMSWTN process without an exhaustion plateau.

7 Gaussian Copula for Correlated Race Finishing Times

In a race model with K accumulators, finishing times $T_1, \ldots, T_K$ are latent random variables representing the hitting times of individual evidence accumulators. Let $F_i(t) = P(T_i \leq t)$ denote the marginal cumulative distribution function (CDF) of accumulator i, $f_i(t) = F_i'(t)$ its marginal probability density function (PDF), and $S_i(t) = 1 - F_i(t)$ its marginal survival function.

When accumulators finish independently, the winning subdensity for accumulator i at time t is $f_i(t) \prod_{j \neq i} S_j(t)$. When finishing times are dependent, their joint distribution can be constructed via a copula.

7.1 Sklar's Theorem and Probit Rank Transformation

By Sklar's theorem, any multivariate joint CDF $F(t_1, \ldots, t_K) = P(T_1 \le t_1, \ldots, T_K \le t_K)$ can be written in terms of its marginal CDFs $F_i(t_i)$ and a copula function $C : [0,1]^K \to [0,1]$:

$$F(t_1, \ldots, t_K) = C\Big(F_1(t_1), \ldots, F_K(t_K)\Big). \tag{36}$$

Under the *Gaussian copula*, each marginal uniform rank variable $U_i = F_i(T_i) \sim U[0,1]$ is mapped to a standard Gaussian random variable using the probit function (inverse standard normal CDF Φ^{-1}):

$$Z_i = \Phi^{-1}(U_i) = \Phi^{-1}\big(F_i(T_i)\big), \qquad i = 1, \ldots, K. \tag{37}$$

The vector $\mathbf{Z} = (Z_1, \ldots, Z_K)^\top$ follows a multivariate normal distribution $\mathcal{N}_K(\mathbf{0}, \mathbf{R})$, where $\mathbf{R}$ is a correlation matrix with unit diagonal and off-diagonal elements $\rho_{ij} = \mathrm{Corr}(Z_i, Z_j)$.

7.2 Bivariate Copula Pair and Winning Subdensities

Consider a race of K accumulators where a single pair (without loss of generality, accumulators 1 and 2) is correlated with copula parameter $\rho \in (-1, 1)$, while all remaining accumulators $j \in \{3, \ldots, K\}$ are mutually independent of each other and of the pair ($\rho_{1j} = \rho_{2j} = 0$).

The latent normal pair (Z_1, Z_2) is standard bivariate normal with correlation ρ:

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right). \tag{38}$$

At time t, define the latent normal quantiles:

$$z_1(t) = \Phi^{-1}\big(F_1(t)\big), \qquad z_2(t) = \Phi^{-1}\big(F_2(t)\big). \tag{39}$$

7.2.1 Subdensity when a Correlated Accumulator Wins

Suppose accumulator 1 wins the race at time t, so $T_1 = t$ and $T_j > t$ for all $j \ge 2$. By independence of accumulators $3, \ldots, K$, the joint winning subdensity is:

$$p_1(t) = f_{T_1, T_2}(t, T_2 > t) \prod_{j=3}^{K} S_j(t). \tag{40}$$

In probit space, the event $T_1 = t$ corresponds to $Z_1 = z_1(t)$ with change-of-variable Jacobian $\mathrm{d}Z_1/\,\mathrm{d}t = f_1(t)/\phi(z_1(t))$, while $T_2 > t$ corresponds to $Z_2 > z_2(t)$. Conditional on $Z_1 = z_1(t)$, the random variable Z_2 is normal with mean $\rho z_1(t)$ and variance $1 - \rho^2$:

$$Z_2 \mid Z_1 = z_1(t) \sim \mathcal{N}\left(\rho z_1(t),\, 1 - \rho^2\right). \tag{41}$$

The conditional survival probability of accumulator 2 given $T_1 = t$ is therefore:

$$P(T_2 > t \mid T_1 = t) = P(Z_2 > z_2(t) \mid Z_1 = z_1(t)) = \Phi\left(\frac{\rho z_1(t) - z_2(t)}{\sqrt{1 - \rho^2}}\right). \tag{42}$$

Multiplying by the marginal density $f_1(t)$, the standard normal PDF factor $\phi(z_1(t))$ from the joint bivariate normal density cancels the denominator of the Jacobian $\mathrm{d}Z_1/\mathrm{d}t$ exactly. The joint density-survival factor is thus:

$$f_{T_1,T_2}(t, T_2 > t) = f_1(t)\,\Phi\!\left(\frac{\rho z_1(t) - z_2(t)}{\sqrt{1-\rho^2}}\right). \tag{43}$$

Consequently, the winning subdensity for accumulator 1 has the exact closed-form expression:

$$p_1(t) = f_1(t)\,\Phi\!\left(\frac{\rho\Phi^{-1}(F_1(t)) - \Phi^{-1}(F_2(t))}{\sqrt{1-\rho^2}}\right)\prod_{j=3}^{K} S_j(t). \tag{44}$$

Equation (44) requires no multivariate integration; it evaluates the winner's marginal density multiplied by a single univariate standard normal CDF applied to the probit-transformed marginal CDFs.

7.2.2 Subdensity when an Independent Accumulator Wins

Suppose an independent accumulator $k \in \{3, \ldots, K\}$ wins the race at time t ($T_k = t$ and $T_j > t$ for all $j \neq k$). The winning subdensity is:

$$p_k(t) = f_k(t)\,P(T_1 > t, T_2 > t)\prod_{j\neq 1,2,k} S_j(t). \tag{45}$$

The joint survival probability of the correlated pair (T_1, T_2) at time t is the upper tail of the standard bivariate normal distribution:

$$P(T_1 > t, T_2 > t) = P(Z_1 > z_1(t), Z_2 > z_2(t)) = \Phi_2(-z_1(t), -z_2(t);\, \rho)\,, \tag{46}$$

where $\Phi_2(x, y; \rho)$ is the standard bivariate normal CDF with correlation ρ. Thus:

$$p_k(t) = f_k(t)\,\Phi_2\big(-\Phi^{-1}(F_1(t)), -\Phi^{-1}(F_2(t));\, \rho\big)\prod_{j\neq 1,2,k} S_j(t). \tag{47}$$

7.2.3 Joint Race Survival

The overall survival function of the race (the probability that no accumulator has finished by time t) is:

$$S_{\text{race}}(t) = P(T_1 > t, \ldots, T_K > t) = \Phi_2\big(-\Phi^{-1}(F_1(t)), -\Phi^{-1}(F_2(t));\, \rho\big)\prod_{j=3}^{K} S_j(t). \tag{48}$$

7.3 Defective Marginals and Intrinsic Non-Finishing

When the drift law retains negative drifts (such as $s_v = 0$ with $m < 0$), the marginal Wald CDF is defective:

$$\lim_{t\to\infty} F_i(t) = p_i = \exp\!\left(\frac{2m_i d_i}{\sigma_i^2}\right) < 1. \tag{49}$$

The marginal timing distribution has a finite plateau at p_i, leaving a point mass $1 - p_i > 0$ at $T_i = \infty$ (intrinsic non-hitting).

The Gaussian copula construction extends cleanly to defective distributions:

- Finite finishing times $T_i < \infty$ map to uniform ranks $U_i \in [0, p_i)$, which correspond to normal quantiles $Z_i \in (-\infty, z_{i,\infty})$, where $z_{i,\infty} = \Phi^{-1}(p_i)$.
- The intrinsic non-finishing event $T_i = \infty$ maps to uniform ranks $U_i \in [p_i, 1]$, which correspond to normal quantiles $Z_i \in [z_{i,\infty}, \infty)$.

Copula uniforms at or above the marginal plateau p_i represent the atom at infinity. Consequently, ρ couples the non-finishing events as well as the finite hitting times:

1. **Joint Omission Probability**: The probability that *neither* accumulator 1 nor 2 ever finishes is given by the bivariate normal upper tail evaluated at the marginal plateaus:

$$P(T_1 = \infty, T_2 = \infty) = P(Z_1 \geq z_{1,\infty}, Z_2 \geq z_{2,\infty}) = \Phi_2\big(-\Phi^{-1}(p_1), -\Phi^{-1}(p_2);\, \rho\big)\,. \tag{50}$$

   For independent defective accumulators ($\rho = 0$), equation (50) reduces to $(1-p_1)(1-p_2)$. Positive correlation ($\rho > 0$) increases the probability of mutual omission.

2. **Partial Omission Probability**: The probability that accumulator 1 finishes by time $t < \infty$ while accumulator 2 never finishes is:

$$P(T_1 \leq t, T_2 = \infty) = P(Z_1 \leq z_1(t), Z_2 \geq z_{2,\infty}) = \Phi(z_1(t)) - \Phi_2(z_1(t), z_{2,\infty};\, \rho)\,. \tag{51}$$

7.4 Non-Parametric Rank Correlation Metrics

The copula correlation parameter $\rho = \mathrm{Corr}(Z_1, Z_2)$ is the latent Gaussian correlation. It maps directly to standard non-parametric rank dependence measures for the pair of finishing times (T_1, T_2):

- **Kendall's τ**:

$$\tau = \frac{2}{\pi} \arcsin(\rho). \tag{52}$$

- **Spearman's ρ_s**:

$$\rho_s = \frac{6}{\pi} \arcsin\left(\frac{\rho}{2}\right)\,. \tag{53}$$

7.5 Finishing-Time Copula vs. Between-Trial Drift Correlation

It is essential to distinguish rank coupling of finishing times from correlation between trial-level drift draws:

1. **Finishing-Time Copula** (Rank Coupling): Couples the operational hitting times T_1, T_2 via Sklar's probit transformation $Z_i = \Phi^{-1}(F_i(T_i))$. It applies directly to the marginal CDFs $F_i(t)$, including when $s_v = 0$ (fixed drift). As shown in equations (44) and (47), the winning subdensity is exact and closed form, requiring only univariate standard normal CDFs and standard bivariate normal CDFs.

2. **Between-Trial Drift Correlation** (Parameter Coupling): Couples the across-trial mean drift rates (V_1, V_2) prior to sampling independent Wald hitting times conditional on (V_1, V_2):
$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} m_1 \\ m_2 \end{pmatrix}, \begin{pmatrix} s_{v1}^2 & \rho s_{v1} s_{v2} \\ \rho s_{v1} s_{v2} & s_{v2}^2 \end{pmatrix}\right). \tag{54}$$
This model requires $s_v > 0$ on the correlated accumulators (when $s_v = 0$, there are no drift draws to correlate). Evaluating the observed race likelihood requires numerical integration (such as Gauss-Hermite quadrature or factor-analytic integration) over the bivariate drift distribution to compute the expected independent race density.

8 Timing Clocks and Outcome Conventions

Let
$$D = t_0 + \tau_d$$
denote the raw decision time and let f_D and S_D denote its subdensity and survival. The survival includes possible non-hit mass when the drift law permits defective Wald processes. Thus $S_D(t) = 1$ for $t < t_0$ and
$$S_D(t) = 1 - F_W(t - t_0; d, \mu, \sigma), \qquad t \geq t_0,$$
in the fixed-drift, fixed-criterion case.

For an Erlang clock with integer shape n and rate λ,
$$f_n(t; \lambda) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \qquad S_n(t; \lambda) = e^{-\lambda t} \sum_{j=0}^{n-1} \frac{(\lambda t)^j}{j!}. \tag{55}$$
Shape 1 is exponential timing:
$$f_1(t; \lambda) = \lambda e^{-\lambda t}, \qquad S_1(t; \lambda) = e^{-\lambda t}.$$
Shape 2 has density and survival
$$f_2(t; \lambda) = \lambda^2 t e^{-\lambda t}, \qquad S_2(t; \lambda) = e^{-\lambda t}(1 + \lambda t).$$
The extra polynomial term in S_2 records the event that a two-phase clock has completed zero or one phase by time t, but not both.

A kill clock and a guess clock enter the race differently. If a kill clock fires before the decision, the trial leaves the observed-response channel. If a guess clock fires before the decision, it generates an observed response from the guessing channel. Consequently the local factors are different:
$$\text{decision observed at } t: \quad f_D(t)\, S_{\text{clock}}(t),$$
$$\text{clock-generated response at } t: \quad f_{\text{clock}}(t)\, S_D(t).$$
The first form is the local-kill form. The second form is the local-guess form.

8.1 Hybrid Erlang Mixture

The hybrid process called shape 3 in this note is a two-component mixture, not an Erlang clock with gamma shape 3. Let $\pi \in [0, 1]$ be the probability of the shape 1 component. If m is the user-facing mean, set $\lambda = 1/m$. The mixture density and survival are

$$f_3(t; \lambda, \pi) = \pi f_1(t; \lambda) + (1 - \pi) f_2(t; 2\lambda), \tag{56}$$

$$S_3(t; \lambda, \pi) = \pi S_1(t; \lambda) + (1 - \pi) S_2(t; 2\lambda). \tag{57}$$

The doubled rate in the shape 2 component puts the two components on the same mean scale:

$$E[T \mid \text{shape } 1] = 1/\lambda = m, \qquad E[T \mid \text{shape } 2] = 2/(2\lambda) = m.$$

For any expression Q that is linear in the clock density or survival,

$$Q_3(\lambda, \pi) = \pi Q_1(\lambda) + (1 - \pi) Q_2(2\lambda). \tag{58}$$

This mixture rule is the canonical way the hybrid clock enters the local kill and local guess derivations below.

9 Laplace Transforms and Wald Tilting

For a nonnegative subdensity f_T, define its Laplace transform by

$$L_T(r) = \int_0^\infty e^{-rt} f_T(t)\,\mathrm{d}t = E[e^{-rT}; T < \infty]. \tag{59}$$

The transform is an exponentially weighted total mass. If a competing clock contributes survival e^{-rt}, then the integral of the observed decision subdensity is exactly $L_T(r)$. Polynomial factors from Erlang clocks are obtained by differentiation:

$$L_T'(r) = -\int_0^\infty t e^{-rt} f_T(t)\,\mathrm{d}t, \qquad L_T''(r) = \int_0^\infty t^2 e^{-rt} f_T(t)\,\mathrm{d}t. \tag{60}$$

For fixed distance and drift, the raw-time transform of $D = t_0 + \tau_d$ is

$$L_D(r) = \exp\left[-rt_0 + \frac{d}{\sigma^2}\left(\mu - \sqrt{\mu^2 + 2\sigma^2 r}\right)\right]. \tag{61}$$

At $r = 0$, this equals the eventual hit probability and therefore remains valid for negative drift.

The same transform appears from an exponential-tilting identity for the Wald density. Define

$$q_r = \sqrt{\mu^2 + 2\sigma^2 r}, \qquad L_\tau(r) = \exp\left[\frac{d}{\sigma^2}(\mu - q_r)\right]. \tag{62}$$

Expanding the Wald exponent gives

$$-\frac{(d - \mu x)^2}{2\sigma^2 x} = -\frac{d^2}{2\sigma^2 x} + \frac{d\mu}{\sigma^2} - \frac{\mu^2 x}{2\sigma^2}.$$

Multiplication by e^{-rx} replaces the coefficient $\mu^2/(2\sigma^2)$ by

$$\frac{\mu^2}{2\sigma^2} + r = \frac{q_r^2}{2\sigma^2}.$$

Therefore

$$f_W(x; d, \mu, \sigma)e^{-rx} = L_\tau(r) f_W(x; d, q_r, \sigma). \tag{63}$$

The drift q_r is an algebraic tilted-drift parameter, not a replacement for the cognitive drift μ.

For survival-weighted clock integrals, define

$$I_0(r) = \int_0^\infty e^{-rt} S_D(t)\,\mathrm{d}t = \frac{1 - L_D(r)}{r}, \tag{64}$$

$$I_1(r) = \int_0^\infty t e^{-rt} S_D(t)\,\mathrm{d}t = \frac{1 - L_D(r)}{r^2} + \frac{L'_D(r)}{r}, \tag{65}$$

$$I_2(r) = \int_0^\infty t^2 e^{-rt} S_D(t)\,\mathrm{d}t = \frac{2(1 - L_D(r))}{r^3} + \frac{2L'_D(r)}{r^2} - \frac{L''_D(r)}{r}. \tag{66}$$

Writing $q_r = \sqrt{\mu^2 + 2\sigma^2 r}$, the derivatives of (61) are

$$L'_D(r) = L_D(r)\left(-t_0 - \frac{d}{q_r}\right), \tag{67}$$

$$L''_D(r) = L_D(r)\left[\left(-t_0 - \frac{d}{q_r}\right)^2 + \frac{d\sigma^2}{q_r^3}\right]. \tag{68}$$

9.1 Finite-Window Tilted Moments

Finite censoring replaces whole-line transforms by tilted moments over the evidence-time window $[0, U]$, where $U = T - t_0$. For a fixed-distance Wald law define

$$M_j(U; r) = \int_0^U x^j f_W(x; d, \mu, \sigma)e^{-rx}\,\mathrm{d}x. \tag{69}$$

Equation (63) gives

$$M_0(U; r) = L_\tau(r) F_W(U; d, q_r, \sigma), \tag{70}$$

$$M_1(U; r) = -\frac{\partial}{\partial r} M_0(U; r), \tag{71}$$

and

$$M_2(U; r) = \frac{d^2}{q_r^2} M_0(U; r) + \frac{\sigma^2}{q_r^2} M_1(U; r) - L_\tau(r)\frac{2d\sigma\sqrt{U}}{q_r^2}\phi\left(\frac{q_r U - d}{\sigma\sqrt{U}}\right). \tag{72}$$

As $U \to \infty$,

$$M_0(\infty; r) = L_\tau(r), \qquad M_1(\infty; r) = \frac{d}{q_r} L_\tau(r), \tag{73}$$

$$M_2(\infty; r) = L_\tau(r)\left(\frac{d^2}{q_r^2} + \frac{d\sigma^2}{q_r^3}\right). \tag{74}$$

Raw-time finite-window decision integrals include the factor e^{-rt_0} and polynomial expansions of $t = t_0 + x$.

10 Local Kill Clock

The local kill process is the simplest competing-risk construction and is the canonical introduction to the Erlang mixture. Let K be independent of D. A response is observed at raw time t when the decision occurs at t and the kill clock has not fired:

$$p_R(t) = f_D(t)S_K(t). \tag{75}$$

This is a subdensity. Its missing mass is assigned to omissions, either because the kill clock fires before the decision or because an upper censoring time is reached before any observed response.

With an upper censoring time T, the observed-response probability is

$$P_R(T) = \int_0^T f_D(t)S_K(t)\,\mathrm{d}t, \tag{76}$$

and the corresponding omission probability is

$$P_O(T) = 1 - P_R(T). \tag{77}$$

Equivalently,

$$P_O(T) = \int_0^T f_K(k)S_D(k)\,\mathrm{d}k + S_K(T)S_D(T), \tag{78}$$

where the first term is kill-before-decision and the second term is unresolved survival to censoring. These two mechanisms are observationally identical when the data code both as omissions.

10.1 Equivalence to Bernoulli Contaminant Misses

At the response-versus-omission level, a local kill clock is exactly a Bernoulli missingness layer. Define

$$p_{\text{miss}}(T) = 1 - P_R(T). \tag{79}$$

Then the trial outcome can be generated by first drawing $C \sim \text{Bernoulli}(p_{\text{miss}}(T))$. If $C = 1$ the trial is coded as an omission; if $C = 0$ an observed response time is drawn from the conditional density

$$f(t \mid R, T) = \frac{f_D(t)S_K(t)}{P_R(T)}, \qquad 0 < t \leq T. \tag{80}$$

This is the same Bernoulli mixture structure as an explicit contaminant-miss parameter: for a finite RT, the likelihood is multiplied by $1-p_{\text{miss}}(T)$, and for an omitted trial the likelihood contribution is $p_{\text{miss}}(T)$ plus any process-omission mass left after the non-contaminant branch.

There is one important qualification. The Bernoulli layer accounts for the *mass* assigned to omissions. The local-kill clock also changes the conditional distribution of observed RTs through the survival factor $S_K(t)$. Thus a killed Wald model is identical to a simple contaminant-miss Wald model only if the observed RT law is also replaced by the killed-and-renormalized law above, or if the analysis uses only the response/omission indicator. If the finite-RT likelihood keeps the original no-kill Wald density, then a free contaminant parameter matches the omission proportion but not the kill-induced RT tilting.

For exponential kill and fixed drift, this conditional RT law is itself another Wald law. From (63),

$$f_W(x; d, \mu, \sigma)e^{-\lambda(t_0+x)} = e^{-\lambda t_0} L_\tau(\lambda) f_W(x; d, q_\lambda, \sigma), \qquad q_\lambda = \sqrt{\mu^2 + 2\sigma^2\lambda}. \tag{81}$$

Therefore exponential local kill with fixed drift can be represented exactly as a Bernoulli miss with

$$p_{\text{miss}}(\infty) = 1 - e^{-\lambda t_0} L_\tau(\lambda),$$

followed, conditional on response, by a Wald with tilted drift q_λ. For Erlang-2 kill the same Bernoulli decomposition still holds, but the conditional RT law is a polynomially tilted Wald,

$$f_W(x; d, \mu, \sigma)e^{-\lambda(t_0+x)}(1 + \lambda(t_0 + x))/P_R,$$

not an ordinary Wald density.

10.2 Exponential Kill

For $K \sim \text{Erlang}(1, \lambda)$,

$$p_R(t) = f_D(t)e^{-\lambda t}. \tag{82}$$

The infinite-window response probability is

$$P_R(\infty) = L_D(\lambda), \tag{83}$$

and the omission probability is $1 - L_D(\lambda)$. For finite $T > t_0$, with $U = T - t_0$,

$$P_R(T) = e^{-\lambda t_0} M_0(U; \lambda). \tag{84}$$

If $T \leq t_0$, then $P_R(T) = 0$ because no decision can occur before non-decision time has elapsed.

10.3 Erlang-2 Kill

For $K \sim \text{Erlang}(2, \lambda)$,

$$p_R(t) = f_D(t)e^{-\lambda t}(1 + \lambda t). \tag{85}$$

The infinite-window response probability is

$$P_R(\infty) = \int_0^\infty f_D(t)e^{-\lambda t}(1 + \lambda t)\,\mathrm{d}t = L_D(\lambda) - \lambda L_D'(\lambda). \tag{86}$$

The derivative term enters because $\int t f_D(t)e^{-\lambda t}\,\mathrm{d}t = -L_D'(\lambda)$. For finite $T > t_0$,

$$P_R(T) = e^{-\lambda t_0}\left[(1 + \lambda t_0)M_0(U; \lambda) + \lambda M_1(U; \lambda)\right]. \tag{87}$$

10.4 Hybrid Kill

For the hybrid kill clock,

$$p_R^{(3)}(t;\lambda,\pi) = \pi f_D(t)S_1(t;\lambda) + (1-\pi)f_D(t)S_2(t;2\lambda). \tag{88}$$

Thus every kill-only response probability is the mixture

$$P_R^{(3)}(T;\lambda,\pi) = \pi P_R^{(1)}(T;\lambda) + (1-\pi)P_R^{(2)}(T;2\lambda), \tag{89}$$

with omission probability $1 - P_R^{(3)}(T;\lambda,\pi)$. The mixture components are both decision-observation subdensities; they differ only in the survival law of the kill clock.

The conditional response-time density given that a response was observed by T is

$$f(t \mid R, T) = \frac{f_D(t)S_K(t)}{P_R(T)}, \qquad 0 < t \le T. \tag{90}$$

This conditional density is distinct from the likelihood subdensity in (75), because the normalization removes the response-versus-omission probability.

11 Local Guess Clock

Let G be a guess clock independent of D. A local guess clock differs from a kill clock because its firing produces an observed response. Therefore the observed-response density has two components:

$$p_R(t) = p_D(t) + p_G(t) = f_D(t)S_G(t) + f_G(t)S_D(t). \tag{91}$$

The first term is a decision response that wins against the guess clock. The second term is a guess response that wins against the decision process. If responses are analyzed without distinguishing their source, the two subdensities are summed.

With upper censoring at T,

$$P_R(T) = \int_0^T f_D(t)S_G(t)\,\mathrm{d}t + \int_0^T f_G(t)S_D(t)\,\mathrm{d}t. \tag{92}$$

Equivalently, because a response has occurred by T unless both the decision process and the guess clock have survived to T,

$$P_R(T) = 1 - S_D(T)S_G(T). \tag{93}$$

The censoring or unresolved probability is $S_D(T)S_G(T)$. In the absence of finite censoring, a proper guess clock eventually fires, so $P_R(\infty) = 1$ even when the decision process is defective.

11.1 Exponential Guess

For $G \sim \text{Erlang}(1, \lambda)$,

$$p_D(t) = f_D(t)e^{-\lambda t}, \qquad p_G(t) = \lambda e^{-\lambda t} S_D(t). \tag{94}$$

The infinite-window component probabilities are

$$P_D(\infty) = L_D(\lambda), \qquad P_G(\infty) = \lambda I_0(\lambda) = 1 - L_D(\lambda). \tag{95}$$

For finite $T > t_0$, the decision component is the same tilted moment as in exponential kill:

$$P_D(T) = e^{-\lambda t_0} M_0(U; \lambda). \tag{96}$$

The guess component can be written either as its defining integral or by complementing the decision component:

$$P_G(T) = \int_0^T \lambda e^{-\lambda t} S_D(t)\,\mathrm{d}t = 1 - e^{-\lambda T} S_D(T) - P_D(T). \tag{97}$$

11.2 Erlang-2 Guess

For $G \sim \text{Erlang}(2, \lambda)$,

$$p_D(t) = f_D(t)e^{-\lambda t}(1 + \lambda t), \qquad p_G(t) = \lambda^2 t e^{-\lambda t} S_D(t). \tag{98}$$

The infinite-window probabilities are

$$P_D(\infty) = L_D(\lambda) - \lambda L_D'(\lambda), \tag{99}$$

$$P_G(\infty) = \lambda^2 I_1(\lambda) = 1 - P_D(\infty). \tag{100}$$

For finite $T > t_0$,

$$P_D(T) = e^{-\lambda t_0} \left[(1 + \lambda t_0) M_0(U; \lambda) + \lambda M_1(U; \lambda)\right], \tag{101}$$

and

$$P_G(T) = 1 - S_2(T; \lambda) S_D(T) - P_D(T). \tag{102}$$

The finite-window complement form is often the most compact expression for the guess component. Direct integration gives the same result after integration by parts.

11.3 Hybrid Guess

For the hybrid guess clock, the observed density is

$$p_R^{(3)}(t; \lambda, \pi) = \pi \left\{f_D(t) S_1(t; \lambda) + f_1(t; \lambda) S_D(t)\right\} + (1 - \pi) \left\{f_D(t) S_2(t; 2\lambda) + f_2(t; 2\lambda) S_D(t)\right\}. \tag{103}$$

Thus

$$P_R^{(3)}(T; \lambda, \pi) = \pi P_R^{(1)}(T; \lambda) + (1 - \pi) P_R^{(2)}(T; 2\lambda). \tag{104}$$

Unlike the kill mixture in (88), each guess mixture component itself contains two observed-response subcomponents: a decision-winning term and a guess-winning term.

12 Combined Local Guess–Kill Clocks

With independent decision, guess, and kill times (D, G, K), the observed-response density is

$$p_R(t) = f_D(t)S_G(t)S_K(t) + f_G(t)S_D(t)S_K(t). \tag{105}$$

The omission channel has a kill component and, under finite censoring, an unresolved component:

$$P_O(T) = \int_0^T f_K(t)S_D(t)S_G(t)\,\mathrm{d}t + S_K(T)S_D(T)S_G(T). \tag{106}$$

For $T = \infty$, the unresolved term vanishes when the kill and guess clocks are proper.

12.1 Exponential Guess and Kill

For exponential guess and kill clocks, set $a = \lambda_g + \lambda_k$. The density is

$$p_R(t) = e^{-at}\left[f_D(t) + \lambda_g S_D(t)\right]. \tag{107}$$

The infinite-window outcome probabilities are

$$p_D = L_D(a), \qquad p_G = \lambda_g I_0(a), \qquad p_O = \lambda_k I_0(a), \tag{108}$$

and $p_D + p_G + p_O = 1$.

For finite T, define

$$D_a(T) = \int_0^T f_D(t)e^{-at}\,\mathrm{d}t. \tag{109}$$

Then

$$P_R(T) = D_a(T) + \lambda_g \frac{1 - e^{-aT}S_D(T) - D_a(T)}{a}. \tag{110}$$

If unresolved trials at T are coded as omissions, the omission probability is $1 - P_R(T)$. The kill-before-censoring component alone is

$$1 - P_R(T) - e^{-aT}S_D(T).$$

12.2 Erlang-2 Guess and Kill

For Erlang-2 guess and kill clocks,

$$S_G(t)S_K(t) = e^{-at}(1 + \lambda_g t)(1 + \lambda_k t), \qquad a = \lambda_g + \lambda_k. \tag{111}$$

The observed-response density is

$$p_R(t) = e^{-at}\left[f_D(t)(1 + \lambda_g t)(1 + \lambda_k t) + \lambda_g^2 t S_D(t)(1 + \lambda_k t)\right]. \tag{112}$$

The infinite-window outcome probabilities are

$$p_D = L_D(a) - aL_D'(a) + \lambda_g\lambda_k L_D''(a), \tag{113}$$

$$p_G = \lambda_g^2 \left[I_1(a) + \lambda_k I_2(a) \right], \qquad p_O = \lambda_k^2 \left[I_1(a) + \lambda_g I_2(a) \right]. \tag{114}$$

For finite $T > t_0$, with $U = T - t_0$, the decision contribution is

$$e^{-at_0} \left[a_0 M_0(U; a) + a_1 M_1(U; a) + a_2 M_2(U; a) \right], \tag{115}$$

where

$$a_0 = 1 + at_0 + \lambda_g \lambda_k t_0^2, \qquad a_1 = a + 2\lambda_g \lambda_k t_0, \qquad a_2 = \lambda_g \lambda_k. \tag{116}$$

The guess contribution is

$$\int_0^T \lambda_g^2 t e^{-\lambda_g t} S_D(t) S_K(t) \, \mathrm{d}t, \tag{117}$$

which reduces to elementary boundary terms and M_0, M_1, M_2 after splitting the integral at t_0 and integrating by parts. This gives closed finite-window Erlang-2 CDFs for fixed-drift Wald models and for uniform start-point variability with no drift variability.

12.3 Hybrid Guess–Kill

If a single shared hybrid component is used for both local timing processes, then

$$p_R^{(3)}(t) = \pi p_R^{(1)}(t; \lambda_g, \lambda_k) + (1 - \pi) p_R^{(2)}(t; 2\lambda_g, 2\lambda_k), \tag{118}$$

and the same mixture rule applies to p_D, p_G, p_O and finite-window CDFs. If the guess and kill clocks draw independent mixture indicators, the expansion has four shape pairs $(1,1), (1,2), (2,1), (2,2)$ with weights $\pi_g \pi_k, \pi_g(1 - \pi_k), (1 - \pi_g)\pi_k, (1 - \pi_g)(1 - \pi_k)$. The shared-component convention is the one assumed in the closed-form boundary below.

13 Why the Contaminant View Does Not Remove the $s_v > 0$ Quadrature

The full-Gaussian no-clock fixed-criterion CDF in (10) is simple because the ordinary Wald CDF contains Gaussian terms that remain Gaussian after integrating over $V \sim \mathcal{N}(m, s_v^2)$.

Erlang timing changes the structure through exponential tilting. Conditional on a drift value v, tilted moments contain

$$q_r(v) = \sqrt{v^2 + 2\sigma^2 r} \tag{119}$$

and factors such as

$$\exp\left[\frac{d}{\sigma^2} \{ v - q_r(v) \} \right] \Phi\left(\frac{q_r(v) U - d}{\sigma \sqrt{U}} \right). \tag{120}$$

The square-root transform $q_r(v)$ prevents completion of the square in v. Consequently, the drift mixture of M_0, M_1, M_2 is not a Gaussian or bivariate-normal integral, even under the full-Gaussian drift law. The simpler expression (10) therefore does not yield a closed-form Erlang-2 local timing CDF when $s_v > 0$.

The no-clock full-Gaussian CDF integrates expressions of the form

$$\Phi(a + bv) \quad \text{and} \quad e^{\alpha v}\Phi(a + bv),$$

which are closed under Gaussian expectation by (12) and (13). Erlang tilting changes these to expressions involving

$$e^{\alpha\{v-\sqrt{v^2+c}\}}\Phi(a + b\sqrt{v^2 + c}),$$

which are not Gaussian expectations of affine probit arguments. This remains true for Erlang-2 because the second-moment formula adds polynomial factors in $1/q_r(v)$ rather than removing the square root.

The Bernoulli-contaminant view does not change this conclusion when the kill parameters are meant to determine the miss probability. It rewrites the model as

$$p_{\text{miss}} = 1 - P_R,$$

but P_R is still the killed response mass. For a fixed criterion and full Gaussian drift,

$$P_R(\infty) = E_V\left[e^{-\lambda t_0}\exp\left\{\frac{d}{\sigma^2}\left(V - \sqrt{V^2 + 2\sigma^2\lambda}\right)\right\}\right] \tag{121}$$

for exponential kill, and Erlang-2 adds the factor

$$1 + \lambda t_0 + \lambda d/\sqrt{V^2 + 2\sigma^2\lambda}$$

inside the same expectation. Equation (121) is already outside the affine-Gaussian probit and Gaussian-tilting identities used earlier. Setting $A = 0$ removes the start-point integral, but it does not remove the square-root transform of the drift, so it does not produce the desired closed form. For $A > 0$, one must also average this non-Gaussian transform over distance.

Using an explicit free $p_{\text{Contaminant}}$ parameter can still be a useful computational alternative, but it changes what is parameterized. It estimates the Bernoulli miss probability directly and no longer provides the deterministic mapping from $(m, s_v, b, A, t_0, \sigma, \lambda)$ to the omission probability implied by an Erlang kill clock. It also leaves the finite-RT law as the no-kill Wald unless the observed-response density is separately replaced by the killed-and-renormalized conditional density. Consequently, the contaminant path is exact for omission-mass modeling, but it is not a closed-form shortcut for the full local-kill likelihood with $s_v > 0$.

For the positive-truncated drift law, the same obstruction remains, with the additional truncation boundary $v > 0$. Thus Erlang-2 timing with $s_v > 0$ is naturally handled by one-dimensional numerical integration over drift or over raw time, depending on the target expression. The hybrid shape 3 process inherits this status: its shape 1 and shape 2 response masses both require the same non-affine drift expectation when $s_v > 0$ and the kill parameter determines the Bernoulli miss probability. The mixture is therefore not generally closed form when the kill component has nonzero weight and $s_v > 0$, although a model that replaces the kill-derived miss mass by a free contaminant parameter avoids that particular calculation by changing the parameterization.

14 Closed-Form Boundary

Case	Density	CDF
Fixed Wald, no clocks	closed form	closed form
Wald with $A > 0$, $s_v = 0$	closed form	closed form
Fixed-criterion full-Gaussian drift, $A = 0$, $s_v > 0$, no clocks	closed form	closed form
Fixed-criterion positive-truncated drift, $A = 0$, $s_v > 0$, no clocks	closed form	closed form
Joint $A > 0$, full-Gaussian drift, no clocks	closed form	one-dimensional start-point average
Joint $A > 0$, positive-truncated drift, no clocks	closed form	one-dimensional start-point average
Gaussian-copula race pair ($s_v = 0$ or positive-truncated marginals)	closed-form sub-density	closed-form survival and bivariate tail
Gaussian-copula race pair with defective marginals ($s_v = 0, m < 0$)	closed-form sub-density	closed-form survival and exact joint omission mass
Correlated between-trial drift draws ($s_v > 0$)	one-dimensional quadrature / factor integration	one-dimensional quadrature / factor integration
Kill-only exponential/Erlang-2, $s_v = 0$	closed form	closed form
Kill-only with $s_v > 0$, kill-derived miss mass	closed-form sub-density in simple density cases	one-dimensional drift/time mixture
Kill-only represented by free Bernoulli contaminant miss	no-kill density unless explicitly replaced by conditional killed density	closed form in the no-kill model; miss mass is a free parameter
Guess-only exponential/Erlang-2, $s_v = 0$	closed form	closed form
Guess-only with $s_v > 0$	closed-form conditional density, drift mixture	drift mixture
Local guess–kill exponential	closed-form density	closed form
Local guess–kill Erlang-2, $s_v = 0$	closed-form density	closed form
Local guess–kill Erlang-2, $s_v > 0$	one-dimensional mixture	one-dimensional mixture
Kill-only or guess-only hybrid shape 3, $s_v = 0$	mixture of shape 1 and 2 closed forms	mixture of shape 1 and 2 closed forms
Local guess–kill hybrid shape 3, $s_v = 0$	mixture of shape 1 and 2 closed forms	mixture of shape 1 and 2 closed forms
Local guess–kill hybrid shape 3, $s_v > 0$	one-dimensional mixture when $\pi < 1$	one-dimensional mixture when $\pi < 1$
GBM physical start-point variability, no drift variability	closed form	closed form

The main rule is structural: closed forms appear when the relevant integrals stay within Gaussian, bivariate-normal, probit copula, or tilted-Wald primitive families. Numerical integration remains appropriate when Erlang tilting introduces $q_r(v) = \sqrt{v^2 + 2\sigma^2 r}$ inside a Gaussian drift mixture, when start-point averaging couples distance and drift in the Wald image term, or when between-trial drift draws are correlated under $s_v > 0$. Hybrid shape 3 does not introduce a new primitive family; it interpolates between the existing shape 1 and shape 2 cases through π. A free Bernoulli contaminant-miss parameter is mathematically equivalent to the killed model's response-versus-omission mass after setting $p_{\text{Contaminant}} = 1 - P_R$, but it is a shortcut only when $p_{\text{Contaminant}}$ is estimated directly or when the killed conditional RT law is supplied separately.

15 Practical Checklist

For learning or deriving a new variant, a useful checklist is:

1. Decide whether negative drifts are retained or the drift distribution is truncated at zero. This determines whether the distribution is defective and what normalizer is legitimate.

2. Decide whether timing clocks use raw time T or evidence time $x = T - t_0$. If clocks start at trial onset, polynomial factors for Erlang-2 contain $t_0 + x$, not just x.

3. Write the target as a density, CDF, survival, or infinite-time mass. Densities are often simpler because they are fixed-time Gaussian moments; CDFs contain reflection-principle image terms.

4. If accumulators in a race are correlated via a Gaussian copula on finishing times, evaluate the probit transformation $z_i(t) = \Phi^{-1}(F_i(t))$. The winning subdensity multiplies the winner's marginal density by a univariate normal CDF $\Phi((\rho z_1 - z_2)/\sqrt{1-\rho^2})$, requiring no multivariate numerical integration.

5. If marginal distributions are defective ($F_i(\infty) < 1$), evaluate the probit quantiles at the marginal plateau $z_{i,\infty} = \Phi^{-1}(F_i(\infty))$ to compute exact joint and partial omission probabilities under the copula.

6. If a clock contributes only e^{-rt}, try a Laplace transform or exponential tilt. If it contributes te^{-rt} or $t^2 e^{-rt}$, use transform derivatives or tilted moments.

7. If the model uses hybrid shape 3 timing, derive the shape 1 and shape 2 quantities first, put the shape 2 rates on the common mean scale by doubling them, and then mix them with weights π and $1 - \pi$.

8. Inspect the remaining random variable integration. Affine Gaussian arguments usually give closed forms; $\sqrt{v^2 + c}$ inside a drift mixture usually does not.

9. If a local kill clock is used only to model omissions, it can be replaced by a Bernoulli contaminant-miss probability. If the kill rate is still meant to determine that probability, the killed response mass P_R must still be computed; with $s_v > 0$ this is the same non-affine drift expectation that motivates quadrature.

16 Further Reading

The formulas above are built from standard components. For deeper theory, start with first-passage times of Brownian motion and the inverse-Gaussian distribution, then study multivariate-normal conditioning and exponential tilting. For timing processes, Erlang distributions are integer-shape gamma distributions; their polynomial-exponential survival functions are why shape 1 and shape 2 are algebraically special. The hybrid shape 3 process is best understood as a finite mixture over those two special cases rather than as a new Erlang distribution. For correlated race models, copulas decompose multivariate distributions via Sklar's theorem; the probit rank transform maps marginal hitting times into Gaussian dependence structures. Useful entry points are the linked references in Section 1, especially inverse Gaussian first-passage theory, the reflection principle, multivariate normal conditioning, copula theory, and Erlang distributions.

For the decision-process context, the supplied sequential-sampling chapter is the reference for the Brownian/Wiener, OU, criterion, and across-trial variability terminology. The supplied time-varying chapter is the reference for filtered sustained and transient drift profiles and for the race-versus- coactivation comparison. Smith (1995) is the primary source for the psychophysically motivated multichannel OU model, while Ratcliff and Van Dongen (2011) provide the one-choice diffusion comparison that motivates the zero-clipping distinction in Section 3.2.